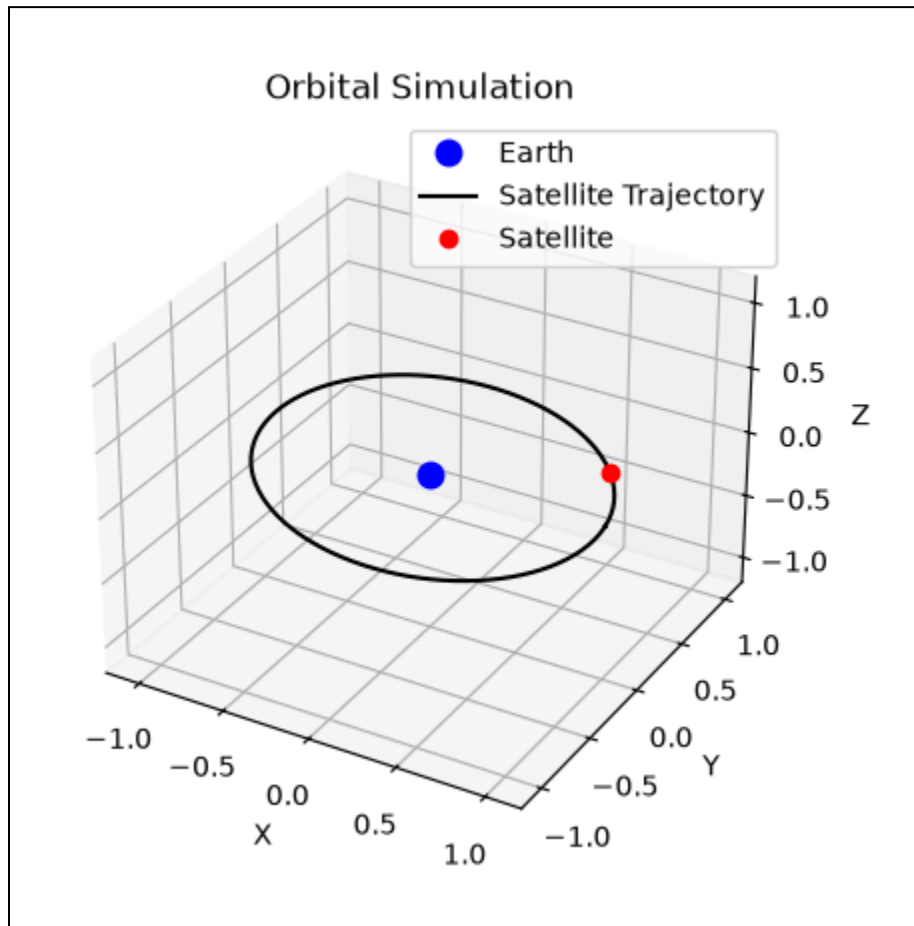


chapter 5 - orbit animation



This script uses NumPy and Matplotlib to simulate and visualize a 3D animation of a satellite orbiting Earth. It generates the elliptical trajectory by using trigonometric sine and cosine functions to calculate a tilted series of x, y, and z coordinates. Matplotlib's FuncAnimation tool then brings the scene to life by continuously updating the position of a red satellite marker along this path frame by frame.